

# Aaroahi M. Singh

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## Professional Summary

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Robotics Engineer specializing in ROS2-based industrial robotic systems, simulation-to-deployment workflows, robot kinematics, motion control, telemetry infrastructure, and autonomous robotics applications. Experienced in developing production-grade robotic software from architecture design through deployment, integrating AC servo systems, industrial communication networks, simulation environments, and real-world robotic hardware. Strong background in AI/ML, reinforcement learning, computer vision, and autonomous systems development for robotics applications.

## Technical Skills

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**Robotics:** ROS2 (Jazzy, Humble), MoveIt2, RViz, Gazebo, URDF, TF2, Forward Kinematics, Inverse Kinematics, Motion Planning, Trajectory Generation, Path Validation, Robot Control Systems

**ROS2 Ecosystem:** ROS2 Packages, Topics, Services, Actions, Lifecycle Nodes, Custom Messages, Custom Services, Hardware Integration, Middleware Design, Deployment Pipelines

**Simulation & Digital Twins:** NVIDIA Isaac Sim, Isaac Lab, Omniverse, PyBullet, MuJoCo, Gymnasium, Physics-Based Simulation, Sensor Modeling, Domain Randomization, Sim-to-Real Validation, Digital Twin Development

**AI/ML for Robotics:** PyTorch, Scikit-Learn, OpenCV, NLTK, Reinforcement Learning, Computer Vision, Perception Pipelines, Sensor Fusion

**Programming:** C++, Python, SQL, Linux (Ubuntu), Git, Docker

**Industrial Automation:** AC Servo Motors, CAN Bus, Telemetry Systems, Remote Monitoring, Robot Diagnostics

## Professional Experience

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### Full Stack Robotics Engineer

Jun 2026 – Present

*P.L. Robotics Pvt. Ltd., Pune*

- Promoted from ROS2 Robotics Development Intern based on exceptional technical performance and ownership of mission-critical robotics projects.
- Leading development of industrial robotics software systems, Delta Robot platforms, and telemetry infrastructure.
- Supporting deployment, migration, integration, monitoring, and maintenance of production robotic systems.

### ROS2 Robotics Development Intern

Dec 2025 – May 2026

*P.L. Robotics Pvt. Ltd., Pune*

- Independently designed and developed a complete ROS2-based Delta Robot platform from architecture design through deployment.
- Implemented Forward Kinematics, Inverse Kinematics, trajectory generation, path validation, motion execution, visualization, and hardware integration layers.
- Developed reusable ROS2 packages, custom interfaces, action servers, services, publishers, subscribers, and middleware components enabling end-to-end robot operation.
- Executed simulation-to-deployment workflows using PyBullet, RViz, custom visualization systems, and physical AC servo hardware.
- Designed software architecture enabling seamless migration from development environments to deployed robotic systems.
- Architected and deployed a ROS2 telemetry platform supporting real-time monitoring, diagnostics, fault analysis, remote operation, ETL pipelines, and SQL-backed analytics.
- Contributed to industrial Cobot software development, CAN-based subsystem integration, robot diagnostics, and automation workflows.

### AI and Robotics Engineer (Contract)

Apr 2025 – Oct 2025

*DRDO, Pune*

- Selected for a Robotics Engineer contract role following successful internship performance and demonstrated technical contributions to robotics research programs.
- Developed reinforcement learning control pipelines for quadruped robotic systems using Gymnasium-based simulation environments.
- Designed robotic task execution frameworks integrating autonomous decision-making and control.
- Developed and validated robotic control policies using Isaac Sim, Isaac Lab, Gymnasium, and PyBullet simulation environments.
- Applied domain randomization and physics parameter tuning to improve sim-to-real transfer performance.
- Validated simulation-trained robotic control policies on physical hardware platforms.

- Developed robotic simulation environments using PyBullet and Gymnasium.
- Optimized contact dynamics, collision handling, and robotic manipulation performance.
- Conducted large-scale simulation experiments for robotic control algorithm evaluation and validation.

Projects

Industrial Delta Robot Platform

- Developed a complete ROS2-based Delta Robot software stack from scratch.
- Implemented FK, IK, trajectory generation, path validation, motion execution, AC servo integration, and simulation environments.
- Built RViz and custom web-based visualization systems for robot monitoring, diagnostics, and debugging.

NVIDIA Isaac Sim Robotics Simulation Platform

- Developed robotics simulation environments using NVIDIA Isaac Sim and Isaac Lab.
- Configured robot dynamics, virtual sensors, collision models, and digital twin workflows.
- Applied domain randomization and sim-to-real validation methodologies for autonomous robotic systems.

Autonomous Hexacopter UAV

- Designed and developed a custom autonomous hexacopter using Pixhawk, ArduPilot, and Raspberry Pi.
- Integrated GPS, IMU, and vision systems for autonomous navigation and flight control.
- Developed simulation environments for validating navigation and perception pipelines before deployment.

Education

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| <b>B.Tech in Electronics and AI &amp; Machine Learning</b> | 2022 – 2025 |
| MIT World Peace University (MIT-WPU), Pune                 | CGPA: 8.79  |
| <b>Diploma in Engineering (ENTC)</b>                       | 2019 – 2022 |
| MIT World Peace University (MIT-WPU), Pune                 | CGPA: 9.53  |